

Rin Liu





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RELEVANT ROLES & EXPERIENCE

ACADEMIC TEACHING & RESEARCH

OCT. 2025 – PRES (CONTRACT)

University of Strathclyde, CIS

Graduate Teaching Assistant

- Assisted in the delivery of undergraduate and postgraduate courses in computer science and mathematics.
- Conducted tutorial sessions and provided support to students in their coursework and research projects.
- Ran labs and tutorials for:
 - CS882 *Security Protocols and Threat Models*, teaching π -calculus and sequent calculus to model concurrent system and formally verify security protocols.
 - CS411 *Theory of Computation*, teaching automata theory, formal grammars and languages, algorithmic complexity theory, and computability theory in the Turing model of computation.
 - CS103 *Machines, Languages, and Computation*, teaching introductory automata theory, logical systems, and proof assistants.

JUN. 2025 – OCT. 2025 (GRANT-FUNDED)

University of Warwick

Undergraduate Researcher

- Awarded a small grant (URSS) to undertake an independent project.
- Developed introductory learning resources on type theory suitable for late-stage undergraduate students.

JUN. 2025 – SEP. 2025 (FREELANCE)

MEI

PSM Academic Mentor

- Trained students in advanced mathematical problem-solving skills.
- Provided tailored academic guidance and mentoring to students.

SEP. 2024 – JUL. 2025 (CONTRACT)

University of Warwick, WMI

Graduate Teaching Assistant

- Led small-group tutorials for core undergraduate maths modules.
- Created targeted exercises and worksheets to reinforce lecture content.

MAR. 2024 – APR. 2025 (STUDENT-LED)

Warwick Maths Society

Academic Support

- Organised and ran weekly mathematics support sessions.
- Delivered large-group lectures for over 300 undergraduate students.
- Developed and taught an introductory course on PGF/TikZ.

PUBLIC ENGAGEMENT & OUTREACH

JUN. 2025 – JUL. 2025 (FULL-TIME, FIXED-TERM)

Good Future Foundation; NatMatSci

Assistant Teacher and Houseparent

- Facilitated interactive lessons for students from diverse backgrounds.
- Assisted in organising and leading educational trips and activities.

EDUCATION

2025 – PRES. Doctor of Philosophy

Mathematically Structured Programming Group

University of Strathclyde

2021 – 2025 MMath (Hons)

FIRST CLASS

Warwick Mathematics Institute

University of Warwick

2019 – 2021 A-Level

A*

Further Mathematics

2019 – 2021 A-Level

A*

Computer Science

2021 Advanced Extension Award

MERIT

2017 – 2019 A-Level

A*

Mathematics

2016 Advanced FSMQ

A

COMPUTER SKILLS

BEGINNER Haskell, OCaml, Java, Prolog, Stenotype

INTERMEDIATE C, Lean 4

PROFICIENT (Cubical) Agda, $\LaTeX$, PGF/TikZ, Python

OTHER SKILLS

LANGUAGE English (Native)

Cantonese (Spoken)

Japanese (Beginner)

MUSIC Violin (Grade 8)

Voice

Piano

JUL. 2024 – JUN. 2025 (FREELANCE)
National Mathematics and Science College
STEM Communication Facilitator
 • Taught research, writing, and presentation skills to advanced students.
 • Guided students through advanced mathematical concepts beyond the standard curriculum.

SEP. 2023 – SEP. 2025 (FREELANCE)
University of Warwick
Mathematics Teaching and Outreach
 • Mentored younger students in mathematics and problem-solving skills.
 • Coordinated and facilitated STEM outreach events and festivals.

AUG. 2023 – SEP. 2025 (VOLUNTEER)
Stand-up Maths Ltd
Branch Organiser
 • Organised the Coventry branch of the **MathsJam** event.
 • Managed contacts and mailing lists, promoting regional STEM events and maths outreach initiatives.

EDITORIAL WORK

MAR. 2024 – DEC. 2024 (STUDENT-LED)
Warwick Maths Society
Publications Officer
 • Independently developed, built, and maintained the **WMS website**.
 • Managed communications within and around the society.

SEP. 2021 – JUL. 2025 (FREELANCE)
Various
Proofreader, Editor, Writer
 • Edited and graphic-designed the *Warwick Poincaré Magazine*.
 • Proofread various textbooks, arXiv preprints, Poincaré articles, etc.

SELECTED WORKS & PROJECTS

Notes on Mathematics

A collection of notes compiled into a reference book, written in $\text{\LaTeX}$, with diagrams written using the $\text{\textit{TikZ}}$ library. (In progress.)

The Yoneda Lemma

A short paper expositing the basics of category theory, culminating in a proof of the celebrated Yoneda Lemma (§5, pp.13-15).

Structural Foundations in Topoi

An analysis of various styles of mathematical foundations, beginning with a discussion of classical material set theories and structuralism, before introducing Lawvere's *Elementary Theory of the Category of Sets*.

102 Combinatorial Problems in Lean 4

A collection of proofs of various olympiad training problems from the book *102 Combinatorial Problems* in the formal proof assistant *Lean 4*, co-authored as part of a three-person team.

Corne Keymap

An ergonomic 40% keyboard firmware for AVR/ARM MCUs, optimised for writing $\text{\LaTeX}$ and C code, with mouse and stenotype protocol emulation.

TALKS

- 2026 FLoC MSFP 2026
Symmetric List Objects
- 2026 MSP101 Seminar
ilts (Symmetric List Objects)
- 2026 TRIPLE 2026
Programming with Cubical Type Theory
- 2026 MSP101 Seminar
Structural Set Theory
- 2025 Warwick-Imperial Mathematics Conference
Structural Set Theory in Foundations
Selected for Outstanding Talk Award
- 2024 Warwick Maths Society Colloquium
Constructivism in Mathematics
- 2024 Undergraduate Seminar Series, WMI
Introduction to Category Theory

REFERENCES

Dr. Siri Chongchitnan

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 Director of Undergraduate Studies
 EMPLOYER Mathematics Institute
University of Warwick
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